



Congressional Stock Trading and Market
Outperformance: Evidence from STOCK Act
Disclosures

Kai Barker Advised by Dr. Tomoyuki Ichiba

December 2026

1 Introduction

Members of Congress have long faced scrutiny over their stock trading activities, with recent polling showing over 80% of Americans support a congressional trading ban(Public Consultation, n.d.). In 1934 the Securities Exchange Act was passed to limit insider trading, but the rules didn't apply to Congress(Brennan Center for Justice, n.d.). The Stop Trading on Congressional Knowledge (STOCK) Act of 2012 attempted to remedy this issue, falling short in practice. The Act mandates that Members disclose trades over 1000 dollars within 45 days and explicitly subjects them to insider trading laws, yet the fine for failing to do so is only 200 dollars(S.2038 – 112th Congress (2011–2012): STOCK Act, n.d.). Members of Congress have access to confidential information, legislative power, knowledge of government contracts, and connections to other influential people. The legality of their trading raises questions, especially given examples of suspicious behavior. Shortly before the Affordable Care Act passed, Senator John Kerry, Senator Jim Webb, Congressman Vern Buchanan, as well as many other politicians, purchased healthcare stocks, with estimated gains between 15% to 50% in the following six-month period(Brennan Center for Justice, n.d.). In another case, the Senate Committee on Health's meeting regarding COVID-19 in January of 2020 led to Senator Dianne Feinstein making significant trades shortly before the market downturn(Brennan Center for Justice, n.d.). Public trust in Congress remains low, with 7 in 10 Americans holding an unfavorable view(Brennan Center for Justice, n.d.). While policies have been proposed to ban Members from acquiring new stocks or require full divestment, no legislation has yet passed. Given the opportunity for profit, examples of possible suspicious trading, and low public sentiment, accusations of wrongdoing are common, but are they justified? Prior research suggests Members of Congress earned significant abnormal returns on their trades before the STOCK Act, but not afterward(Huang and Xuan, n.d.). Other studies have found that stock ownership correlates with politicians making decisions favorable to those respective companies, suggesting a mutual interest from Members and corporate stakeholders(Tahoun, n.d.). However, these studies were both published nearly a decade ago and don't account for new data, especially the over \$150 million traded by Congress during the COVID-19 pandemic(Brennan Center for Justice, n.d.). With 12 years of STOCK Act disclosures now available from 2012 to 2024, this study examines whether U.S. Congressional Members generate abnormal returns on their stock trades, analyzing overall portfolio performance, and whether committee membership generates superior overall returns.

2 Data Cleaning and Preprocessing

For this study, historical time series data was needed for three components. The first component was the Members and their trades, where publicly available data can be collected from disclosures on the official government websites, however the format of the data (PDFs or images of disclosures) made it incred-

ibly difficult to retrieve. Instead an ensemble of three datasets, sourced from Kaggle, provided a comprehensive collection of congressional trades (~70,000 rows). These datasets spanned different ranges between 2012 and 2024, and contained 46,000, 15,000, and 5,000 observations respectively, with many duplicates. However, a limitation with the data sources was the verification of trades with those available on the government websites, as well as differing amounts of information contained, and most notably, naming conventions such as including suffixes or nicknames. The second component was the prices of the stocks themselves, needed for determining profits and losses over holding periods. This data was also sourced from Kaggle as well as the yfinance API. The third component was our benchmark, the S&P 500. This was selected as a baseline due to its historic use as an indication of overall market health, broad scope of sectors, and market capitalization weighting(Reiff, n.d.). We assume that all the collected data is accurate, and we base our analysis on the closing prices for stocks and the S&P 500.

In order to clean the data, sale and purchase amounts are disclosed as ranges, though some of the datasets only included one bound. These needed to be converted to the disclosure ranges that separate into “1,001–15,000”, “15,001–50,000”, and so on up to “Over 50,000,000”. Since our analysis only focuses on profits, in percentages, the actual purchase amounts could be neglected, but the ranges could be useful for further analysis. To deal with the issue of name mismatches, nicknames were standardized, and all names were reformatted to “First, Last”, removing suffixes and occasionally middle names if possible. Over 2,000 tickers failed to download from yfinance and needed to be sourced from the Kaggle dataset or cleaned, with some companies being currently delisted, acquired, or referencing assets from foreign markets. On that point, it is important to note that we are only examining stocks traded on the U.S. market and ETFs, bonds, and foreign assets were not considered. After all the previous cleaning and combining, any persisting duplicates were removed, and purchases and sales of stocks based on senator were paired together (using First In First Out). This left a final dataset of 7,514 observations, consisting of a purchase and sale of a stock, a holding period, and profit percentage calculated for the duration it stayed in a Members portfolio.

3 Data Analysis

First, we returns from the S&P 500 were calculated over the same time period that the stock trades cover, and the compound annual growth rate was calculated to be 12.49

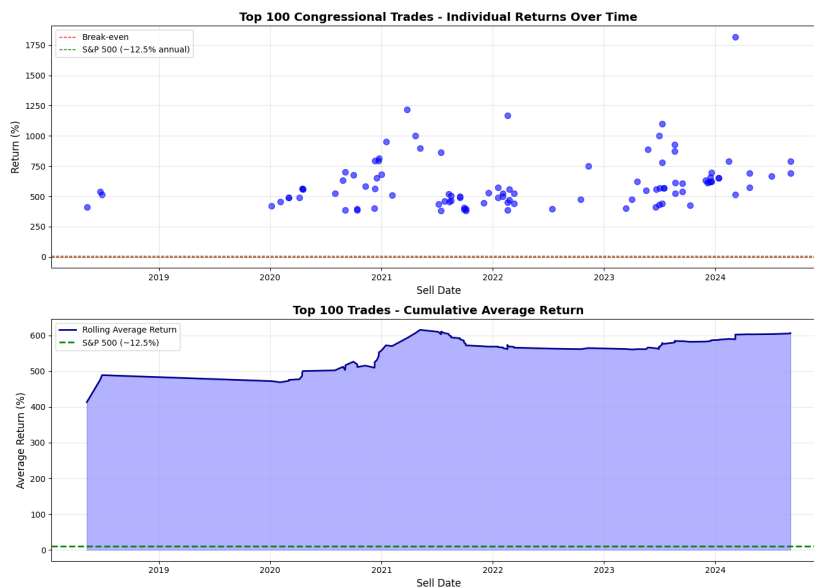


Figure 1: Figure 1 was created to visualize the Top 100 Congressional Trades, based on the profit percentage. We can see that all of these have over 250% returns. Note that this is just a visualization, and does not have statistical significance.

For each trade, excess returns were calculated by comparing the stock's performance over the holding period to the S&P 500's performance over the same period.

Metric	Value
Number of trades analyzed	7,509
Total trades	7,510
Mean Profit (%)	+32.96
Mean excess return (%)	+13.17
Median excess return (%)	+0.34
Std Dev excess return (%)	81.56
IQR excess return (%)	38.72
25th percentile (%)	-17.82
75th percentile (%)	+20.91
Min excess return (%)	-155.97
Max excess return (%)	+1,702.18
% of trades beating S&P 500	50.7
Wilcoxon test statistic	14,915,012.00
Wilcoxon p-value	0.000007
Significance	***

4 add shapiro table

5 References

References

- Brennan Center for Justice. n.d. “Congressional Stock Trading, Explained.” Accessed December 1, 2024. <https://www.brennancenter.org/our-work/research-reports/congressional-stock-trading-explained>.
- Huang, Ruidi, and Yuhai Xuan. n.d. ““Trading” Political Favors: Evidence from the Impact of the STOCK Act.” *SSRN Electronic Journal*, accessed January 1, 2020. https://business.gwu.edu/sites/g/files/zaxdzs5326/files/HuangXuan_STOCKAct.pdf.
- Public Consultation. n.d. “Ban on Stock Trading for Members of Congress Favored by Overwhelming Bipartisan Majority.” Accessed December 1, 2024. <https://publicconsultation.org/united-states/stock-trading-by-members-of-congress/>.
- Reiff, Nathan. n.d. “Why Do Investors Use the S&P 500 as a Benchmark?” Investopedia. Accessed December 1, 2024. <https://www.investopedia.com/ask/answers/041315/what-are-pros-and-cons-using-sp-500-benchmark.asp>.
- Tahoun, Ahmed. n.d. “The Role of Stock Ownership by US Members of Congress on the Market for Political Favors.” *Journal of Financial Economics*, accessed January 1, 2014. <https://www.sciencedirect.com/science/article/abs/pii/S0304405X13002705>.